

Exercise 1: Configuring a Basic Spring Application

BookRepository.java:

```
package com.exercise1.repository;
```

```
public class BookRepository {  
    public void saveData() {  
        System.out.println("[BookRepository] Book data saved to the database successfully!");  
    }  
}
```

BookService.java:

```
package com.exercise1.service;
```

```
import com.exercise1.repository.BookRepository;
```

```
public class BookService {  
    private final BookRepository bookRepository;  
  
    // Constructor Injection for Spring container wiring  
    public BookService(BookRepository bookRepository) {  
        this.bookRepository = bookRepository;  
    }  
  
    public void saveBook() {  
        System.out.println("[BookService] Processing book business logic...");  
        bookRepository.saveData();  
    }  
}
```

```
}  
}
```

applicationContext.xml:

```
<?xml version="1.0" encoding="UTF-8"?>  
  
<beans xmlns="http://www.springframework.org/schema/beans"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="http://www.springframework.org/schema/beans  
        http://www.springframework.org/schema/beans/spring-beans.xsd">  
  
    <bean id="bookRepository" class="com.exercise1.repository.BookRepository" />  
  
    <bean id="bookService" class="com.exercise1.service.BookService">  
        <constructor-arg ref="bookRepository" />  
    </bean>  
  
</beans>
```

App.java:

```
package com.exercise1;  
  
// 1. Import the Spring classes  
import org.springframework.context.support.ClassPathXmlApplicationContext;  
  
// 2. Import your custom service class from the other package  
import com.exercise1.service.BookService;  
  
public class App {
```

```
public static void main(String[] args) {  
    // Load the XML application context configuration  
    ClassPathXmlApplicationContext context = new  
ClassPathXmlApplicationContext("applicationContext.xml");  
  
    // Retrieve the configured BookService bean  
    BookService bookService = context.getBean("bookService", BookService.class);  
  
    // Test the setup  
    bookService.saveBook();  
  
    // Close the context context gracefully  
    context.close();  
}  
}
```

Output:

```
● PS D:\SD\Digital-Nurture-JavaFSE-main\local repo> & 'C:\Program F  
cal\Temp\cp_bfb168c90a607d71inxsdjesz.argfile' 'com.exercise1.App'  
[BookService] Processing book business logic...  
[BookRepository] Book data saved to the database successfully!
```